

Extended-state Cartan telescope, causal Doob one-use, and rational recovery boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-098 left three closely related questions at the A13 frontier. Could a signed Cartan visit be grouped when its target heat changes? Could the bare future-control response be assigned once without differentiating an adapted control? Could the exact R-085 rational recovery be completed once an upper, rather than merely lower, estimate for its five unshifted families is known?

This note answers all three questions at their honest scope and records two route exclusions.

1. A complete Cartan edge in the extended state (Σ, z) telescopes exactly within one fixed Malliavin root. The heat-change bracket is mandatory. Any common linear root derivative, Ornstein–Uhlenbeck operator, and FAR projection preserve the already-grouped identity.
2. A later inverse visit carried by a fresh temporal root does not cancel the earlier nonnegative root square. A production Cartan harmonic therefore rules out a cutoff/partition/revisit-uniform extension of the R-085 mixed-only ledger to arbitrary progressive controls. The result does not touch R-092’s regular no-revisit theorem and does not exclude a once-only pure- X allowance.
3. Chronological Doob differences give an exact derivative-free Doob–Hardy theorem. Each Hoeffding component is owned at its last revealed root and all later smoothing shells are grouped before squaring. This improves ownership and signed grouping; it does not invalidate the already-correct weighted resampling estimate of R-098.
4. The complete nonlinear frame has an exact ordered-reveal decomposition into a mean secant and a Jensen residual. A finite product fixture proves that the residual cannot be controlled by the same-level shift increment alone, and a rare-event spike rules out squaring the spatial frame secant before its signed cross pairing.
5. The five R-085 unshifted rational families satisfy a two-sided form bound on the accepted fixed-floor regular strict-past one-shot class. An exact payment-gauge identity then reduces rational (6.5) to a lower form for the complete owner packet, which is weaker than positivity of the posterior bracket alone. The latter lower form is still open.

The result is a T4 analytic/exact/executed advance and method boundary. It does not prove the coefficient-unconditioned production posterior lower form, rational (6.5), complete H_N , REG, arbitrary progressive H_A , OVERLAP_{src}, the $q = 10/9$ Nelson estimate, cutoff or floor removal, an interacting measure, or Sector-A closure.

2 Authorities, notation, and scope

The production Cartan current is based on a trace-free Cartan matrix S and the quotient vector

$$F_S(z) = \frac{z^T S z}{e + |z|^2} z, \quad J_\Sigma(z) = (P_\Sigma F_S)(z)^T D z, \quad (2.1)$$

where $e > 0$ is the fixed floor and P_Σ is the Gaussian heat operator with target covariance Σ . Formula (2.1) is a one-form; D_z below denotes a value derivative and D_x a spatial derivative.

R-083 supplies the exact raw value/heat telescope, R-084 the root-diagonal Cartan current and CFAR normalization, R-085 the targets (4.10)–(4.11) and (6.5), R-087 the accepted regular spatial decay, R-092 the regular mutually-orthogonal strict-past no-revisit Cartan theorem, R-097 the terminal Gram/posterior and cross-Doob owners, and R-098 the exact production harmonic, frame secant, rational ridge, payment split, and same-heat reverse refinement. Every use below is at finite cutoff and fixed positive floor.

For the Doob theorem, $(\mathcal{F}_j)_{j=j_0-1}^J$ is the chronological filtration of independent roots. All operators K_k are deterministic, and $h_k \in L^2(\mathcal{F}_{k-1}; H)$. The base conditional mean is $\mathbb{E}_{j_0-1} h_k$, not necessarily the unconditional mean.

3 The extended-state signed Cartan telescope

Fix one Malliavin root j and a finite path $(\Sigma_\ell, z_\ell)_{\ell=0}^L$. Define the complete edge

$$\begin{aligned} E_\ell &= J_{\Sigma_\ell}(z_\ell) - J_{\Sigma_\ell}(z_{\ell-1}) \\ &\quad + J_{\Sigma_\ell}(z_{\ell-1}) - J_{\Sigma_{\ell-1}}(z_{\ell-1}). \end{aligned} \quad (3.1)$$

The first line is the value/source edge at the new heat and the second line is the target-heat compensator.

Theorem 3.1 (complete common-root telescope)

For every finite path, without a sign or monotonicity assumption,

$$\boxed{\sum_{\ell=1}^L E_\ell = J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0)}. \quad (3.2)$$

Consequently, for a common root derivative, OU time, and FAR projector,

$$Q_{j,C} P_t^{(j)} \mathbb{D}_j \sum_{\ell=1}^L E_\ell = Q_{j,C} P_t^{(j)} \mathbb{D}_j \{J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0)\}. \quad (3.3)$$

Proof. Each complete edge is already $J_{\Sigma_\ell}(z_\ell) - J_{\Sigma_{\ell-1}}(z_{\ell-1})$. Summation gives (3.2). Equation (3.3) follows by applying the same linear operator to the grouped identity. $\square$

Thus arbitrary signed subdivisions inside one root are partition invariant. Equality of both extended endpoints guarantees cancellation, but it is not necessary because $(\Sigma, z) \mapsto J_\Sigma(z)$ need not be injective; for example, distinct heats agree at $z = 0$. Equality of the value endpoints alone does not guarantee cancellation, and a heat change generically leaves the defect in (3.4). At fixed heat the compensators vanish and Theorem 3.1 reduces to the R-098 same-root reverse cancellation. Identifying the endpoint derivative with the accepted R-084 three-channel formula also requires the endpoint shift to be $\mathcal{F}_{j-1}$ -measurable; otherwise $\mathbb{D}_j a$ is a new channel.

Omitted-heat defect

If the heat compensators are dropped, the source-only sum is

$$S_{\text{src}} = J_{\Sigma_L}(z_L) - J_{\Sigma_0}(z_0) - \sum_{\ell=1}^L [J_{\Sigma_\ell} - J_{\Sigma_{\ell-1}}](z_{\ell-1}). \quad (3.4)$$

The last line is an exact defect, not an error term.

For the production vector F_S with $\text{tr } S = 0$, its cubic Taylor term gives

$$D_z(\Delta F_S)(0) = \frac{4S}{e}, \quad D_z(P_{\tau_I} F_S)(0) = \frac{2\tau S}{e} + O(\tau^2). \quad (3.5)$$

Indeed, $\Delta[(z^T S z)z] = 2(\text{tr } S)z + 4Sz$. For $f_\epsilon = (\sin \theta, \epsilon \cos \theta)$, $z = \delta f_\epsilon$, and a root direction $v = \sqrt{\lambda} f_\epsilon$, the leading omitted-heat derivative is

$$\frac{2(\tau_1 - \tau_2)}{e} \delta \sqrt{\lambda} D_x(f_\epsilon^T S f_\epsilon) + O\{\delta(\tau_1^2 + \tau_2^2) + \delta^3\}. \quad (3.6)$$

For the selected Cartan harmonic, $D_x(f_\epsilon^T S f_\epsilon)$ is a nonzero multiple of $(1 + \epsilon^2) \sin(2\theta)$. Hence (3.4) is nonzero in the production current. This low-harmonic diagnostic proves that deleting the heat owner is false; it is not by itself a FAR counterexample.

4 Progressive revisits: the terminal mixed-only extension is false

Theorem 3.1 does not combine distinct temporal roots. Consider an admissible finite temporal chart in the repeated-range class already used in R-094. Before a selected root j , insert the deterministic predictable shift

$$a_j = A f_\epsilon. \quad (4.1)$$

At a later temporal packet, insert $-A f_\epsilon$. The terminal shift is $A^* = 0$, but the first root still sees $A^{(j)} = A f_\epsilon$.

R-098 computes a nonzero output harmonic of the actual forward production operator. For every fixed $C \geq 5$, choose one nonzero output m ; then for all sufficiently large A ,

$$\int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(j)} \mathbb{D}_j H_{A,j,A f_\epsilon}\|_{\text{HS}}^2 dt \geq c_{C,\epsilon,n_j} A^2, \quad (4.2)$$

where $c_{C,\epsilon,n_j} > 0$. Before the fixed LP transfer, its squared coefficient contains

$$N_j^2 \lambda_{n_j} \epsilon^2 (1 - \rho^2)^2 \rho^{2(k_C-1)}, \quad \rho = \frac{1 - \epsilon}{1 + \epsilon}. \quad (4.3)$$

The R-084 CFAR normalization therefore gives

$$S_C^{\text{ctrl}} \geq \frac{3}{40P} c_{C,\epsilon,n_j} A^2. \quad (4.4)$$

On this loop, the source cost is $X = \kappa_{\text{loop}} A^2$ and $Y = \|U + A^*\|_6^6 = \|U\|_6^6$ is independent of A . Hence

$$1 + \mathbb{E}[X^{1/2} Y^{1/2}] = O(1 + A), \quad (4.5)$$

which cannot dominate (4.4) uniformly as $A \rightarrow \infty$.

Theorem 4.1 (scoped progressive mixed-payload no-go)

There is no cutoff/partition/revisit-uniform extension of the R-085 (4.11)-type ledger to arbitrary progressive revisits if its only growing right-hand payload is the terminal mixed quantity $\mathbb{E}[X^{1/2}Y^{1/2}]$.

The proof is (4.1)–(4.5). It needs only the nonnegative first-root square; no identification of the later root operator with the first is required. The later inverse visit cannot delete a previously diagonalized root square.

The theorem has four important exclusions. It does not refute R-092’s regular no-revisit H_C . It does not exclude an ηX allowance; on this loop such an estimate needs a coefficient at least

$$\eta \geq \frac{3c_C}{40P\kappa_{\text{loop}}}. \quad (4.6)$$

It is not a counterexample to the complete source action or the R-097 posterior packet, which retain the $9X/20$ action term and signed heat, low, future, rational, trace, and forest companions. Finally it does not create a new positive reserve: terminal $A^* = 0$ makes the terminal Gram difference zero while the intermediate root square remains.

The method failure is registered as [NG-2026-07-27-A13-PROGRESSIVE-REVISIT-CARTAN-MIXED-PAYLOAD](#).

5 Causal Doob–Hardy ownership

Let

$$A^* = \sum_{k=j_0}^{J+1} K_k h_k, \quad d_j = \mathbb{E}[\cdot \mid \mathcal{F}_j] - \mathbb{E}[\cdot \mid \mathcal{F}_{j-1}], \quad (5.1)$$

with deterministic $K_k : H \rightarrow V$, $h_k \in L^2(\mathcal{F}_{k-1}; H)$, and $\|K_k\| \leq \kappa 2^{-\alpha k}$.

Theorem 5.1 (last-root Doob–Hardy theorem)

For every $\beta > 0$, set $S_\beta = \sum_{r \geq 1} 2^{-\beta r} = (2^\beta - 1)^{-1}$. Then

$$d_j A^* = \sum_{k > j} K_k d_j h_k \quad (5.2)$$

and

$$\boxed{\sum_j 2^{\beta j} \mathbb{E} \|d_j A^*\|_V^2 \leq S_\beta \kappa^2 \sum_k 2^{-(2\alpha-\beta)k} \sum_{j < k} \mathbb{E} \|d_j h_k\|_H^2.} \quad (5.3)$$

Consequently,

$$\boxed{\sum_j 2^{\beta j} \mathbb{E} \|d_j A^*\|_V^2 \leq S_\beta \kappa^2 \sum_k 2^{-(2\alpha-\beta)k} \mathbb{E} \|h_k - \mathbb{E}_{j_0-1} h_k\|_H^2.} \quad (5.4)$$

Proof. If $k \leq j$, then h_k is already $\mathcal{F}_{j-1}$ -measurable, proving (5.2). For fixed j , weighted Cauchy–Schwarz gives

$$\left\| \sum_{k > j} K_k d_j h_k \right\|^2 \leq \kappa^2 \left(\sum_{k > j} 2^{-\beta(k-j)} \right) \sum_{k > j} 2^{\beta(k-j)} 2^{-2\alpha k} \|d_j h_k\|^2.$$

Multiplying by $2^{\beta j}$ and summing proves (5.3). Martingale orthogonality for each h_k proves (5.4). $\square$

For the production L^2 smoothing exponent $\alpha = 2$ and $\beta = 1$, $S_1 = 1$, so

$$\sum_j 2^j \mathbb{E} \|d_j A^*\|_2^2 \leq \kappa_K^2 \sum_k 2^{-3k} \mathbb{E} \|h_k - \mathbb{E}_{j_0-1} h_k\|_2^2. \quad (5.5)$$

The constant one is sharp in the limiting finite-array sense. In three spatial dimensions, the accepted estimate

$$\|K_k\|_{L^2 \rightarrow L^r} \lesssim 2^{-(1/2+3/r)k}, \quad 2 \leq r \leq 6, \quad (5.6)$$

gives

$$\sum_j 2^j \mathbb{E} \|d_j A^*\|_r^2 \lesssim \sum_k 2^{-(6/r)k} \mathbb{E} \|h_k - \mathbb{E}_{j_0-1} h_k\|_2^2. \quad (5.7)$$

The decay powers at $r = 2, 3, 6$ are 3, 2, 1.

Exact Hoeffding accounting

Write the orthogonal Hoeffding expansion over roots before k as

$$h_k = \sum_{S \subset \{j_0, \dots, k-1\}} h_{k,S}. \quad (5.8)$$

Chronological revelation yields

$$d_j h_k = \sum_{\max S=j} h_{k,S}; \quad (5.9)$$

each nonconstant component is owned exactly once. Coordinate resampling instead satisfies

$$\frac{1}{2} \mathbb{E} \|h_k - h_k^{(j)}\|^2 = \sum_{S \ni j} \|h_{k,S}\|_2^2, \quad (5.10)$$

so its unweighted accounting sees a component $|S|$ times. This is not a defect in R-098: the existing dyadic weights obey $\sum_{j \in S} 2^j < 2^k$ and retain its 2^{-3k} conclusion. The new gain is chronological ownership, removal of the resampling factor, and signed grouping of every future k before squaring.

6 The complete-frame ordered reveal

Let Z be the terminal value field, assume $B_{\text{fr}}(Z) \in L^2$, and define

$$Z_j = \mathbb{E}_j Z, \quad \widehat{B}_j = \mathbb{E}_j B_{\text{fr}}(Z), \quad \mathcal{J}_j = \widehat{B}_j - B_{\text{fr}}(Z_j). \quad (6.1)$$

Theorem 6.1 (mean secant plus Jensen residual)

For every reveal,

$$\boxed{d_j \widehat{B} = B_{\text{fr}}(Z_j) - B_{\text{fr}}(Z_{j-1}) + \mathcal{J}_j - \mathcal{J}_{j-1}.} \quad (6.2)$$

If Q_j is the R-097 quadratic-Wick martingale, then for any base ℓ ,

$$\begin{aligned} \mathbb{E} \langle B_{\text{fr}}(Z), Q_J \rangle &= \mathbb{E} \langle \widehat{B}_\ell, Q_\ell \rangle \\ &\quad + \sum_{j>\ell} \mathbb{E} \langle B_{\text{fr}}(Z_j) - B_{\text{fr}}(Z_{j-1}) + \mathcal{J}_j - \mathcal{J}_{j-1}, d_j Q \rangle. \end{aligned} \quad (6.3)$$

Proof. Equation (6.2) follows by adding and subtracting the two mean frames. Equation (6.3) is martingale orthogonality. $\square$

The decomposition is invariant under subdivisions that leave both the terminal field Z and the filtration/reveal order unchanged. It cannot be declared invariant after adding fresh roots or changing their order. The global Schur mean

$$-q^T (\overline{B} + 2R)^{-1} q \quad (6.4)$$

also remains one unsplit owner; one must not condition on the full terminal control and then reuse the original root independence.

For the quadratic Gram $B(z) = z^{\otimes 2}$, $\mathcal{J}_j = \text{Cov}(Z \mid \mathcal{F}_j)$. With $u_j = d_j Z$,

$$\begin{aligned} d_j \hat{B} &= Z_{j-1} \otimes u_j + u_j \otimes Z_{j-1} + u_j \otimes u_j \\ &\quad + \text{Cov}(Z \mid \mathcal{F}_j) - \text{Cov}(Z \mid \mathcal{F}_{j-1}). \end{aligned} \quad (6.5)$$

The last line is a mandatory Jensen/covariance residual. It is structurally the kind of residual the posterior/forest architecture must own, but this note does not identify it literally with the R-097 J_B or the R-063 forest.

A finite exponential separation

Let independent ξ_r have

$$\Pr(\xi_r = \pm 2) = \frac{1}{8}, \quad \Pr(\xi_r = 0) = \frac{3}{4}. \quad (6.6)$$

Then $\mathbb{E}\xi_r = 0$, $\mathbb{E}\xi_r^2 = 1$, and $\mathbb{E}\xi_r^4 = 4$. For $Z = \prod_{r=1}^n \xi_r$, one has $d_j Z = 0$ for $j < n$ and $d_n Z = Z$, while

$$d_j \hat{B} = \left(\prod_{r < j} \xi_r^2 \right) (\xi_j^2 - 1), \quad \mathbb{E}|d_j \hat{B}|^2 = 3 \cdot 4^{j-1}. \quad (6.7)$$

Therefore

$$\sum_{j=1}^n \mathbb{E}|d_j \hat{B}|^2 = 4^n - 1, \quad \sum_{j=1}^n \mathbb{E}|d_j Z|^2 = 1. \quad (6.8)$$

No cutoff-uniform nonlinear-frame estimate can use only the same-level mean shift $d_j Z$.

Why the frame secant must remain linearly paired

R-098 gives

$$\|B_{\text{fr}}(z) - B_{\text{fr}}(w)\|_F \leq C_B(|z| + |w|)|z - w|. \quad (6.9)$$

Thus

$$\|B_{\text{fr}}(Z_j) - B_{\text{fr}}(Z_{j-1})\|_{L_x^2} \leq C_B(\|Z_j\|_6 + \|Z_{j-1}\|_6)\|d_j Z\|_3. \quad (6.10)$$

Keeping (6.10) linearly paired with $d_j Q$ has formal payload $X^{1/2}Y^{1/6}$ and total exponent $2/3 < 1$. Squaring first creates $XY^{1/3}$ and total exponent $4/3 > 1$.

This loss is real. On an event E with probability N^{-6} set

$$u = N^3 \mathbf{1}_E, \quad z = N \mathbf{1}_E. \quad (6.11)$$

Then

$$\mathbb{E}u^2 = \mathbb{E}z^6 = \mathbb{E}(|u|z^3) = 1, \quad \mathbb{E}(z^2 u^2) = N^2. \quad (6.12)$$

Equations (6.8) and (6.12) register the method failure [NG-2026-07-27-A13-ABSOLUTE-LAST-ROOT-FRAME-TRANSFER](#). They do not refute a signed linear estimate for the complete posterior packet.

7 Rational five-family two-sided form

On the fixed-floor regular mutually-orthogonal strict-past one-shot class, R-084 and R-085 give absolute pathwise estimates for the five unshifted families

$$\begin{aligned} U_1 &= \frac{1}{2}D\overline{B}(U)[a] : Q, & U_2 &= \frac{1}{4}D^2\overline{B}(U)[a, a] : Q, & U_3 &= G^T\overline{B}(U)c, \\ U_4 &= G^TD\overline{B}(U)[a]c, & U_5 &= \frac{1}{2}G^TD^2\overline{B}(U)[a, a]c. \end{aligned} \quad (7.1)$$

Writing $X = \|a\|_{H^2}^2$, $Y = \|U + a\|_6^6$, they have the form

$$|U_\ell| \leq C_\ell M_\ell X^{x_\ell} Y^{y_\ell}, \quad (7.2)$$

with the exact exponent ledger

ℓ	(x_ℓ, y_ℓ)	$\sigma_\ell = 1 - x_\ell - y_\ell$	$p_\ell = \sigma_\ell^{-1}$
1	(11/40, 3/40)	13/20	20/13
2	(11/20, 3/20)	3/10	10/3
3	(2/5, 1/30)	17/30	30/17
4	(2/5, 1/5)	2/5	5/2
5	(2/5, 11/30)	7/30	30/7.

(7.3)

If $a + b + \sigma = 1$, weighted Young gives

$$\begin{aligned} CMX^a Y^b &\leq \eta X + \zeta Y \\ &\quad + \sigma C^{1/\sigma} a^{a/\sigma} b^{b/\sigma} \eta^{-a/\sigma} \zeta^{-b/\sigma} M^{1/\sigma}. \end{aligned} \quad (7.4)$$

Allocate $\eta/5$ and $\zeta/5$ to each family. Every required moment is at most 30/7, which is available in the accepted reconstructed model theory.

Theorem 7.1 (two-sided unshifted form)

For every $\eta, \zeta > 0$ in the declared regular one-shot class,

$$\boxed{\mathbb{E}|\mathcal{U}_R| \leq \eta \mathbb{E}X + \zeta \mathbb{E}Y + C_{\eta, \zeta, e, j_0}.} \quad (7.5)$$

In particular, $\mathcal{U}_R$ has both lower and upper form bounds. The result uses no independence of the adapted test a , but it is not a theorem for arbitrary progressive/revisit controls.

8 Payment gauge and the exact remaining rational owner

On one conditional fibre, set

$$K = \mathbb{E}_{\mathcal{F}}B, \quad q = \mathbb{E}_{\mathcal{F}}(BG), \quad A = K + 2R, \quad (8.1)$$

where $B = B^T \succeq 0$, $R \succ 0$, and c, R, Γ are $\mathcal{F}$ -measurable. Assume all displayed terms are integrable.

Theorem 8.1 (payment-gauge identity)

$$\boxed{\begin{aligned} &\frac{1}{2}\|c + A^{-1}q\|_A^2 + \frac{1}{2}\{\mathbb{E}_{\mathcal{F}}[B : (G^{\otimes 2} - \Gamma)] - q^T A^{-1}q\} - c^T R c \\ &= \frac{1}{2}\mathbb{E}_{\mathcal{F}}[B : ((G + c)^{\otimes 2} - \Gamma)]. \end{aligned}} \quad (8.2)$$

Proof. Expanding the first square cancels $q^T A^{-1}q$, while

$$\frac{1}{2}c^T(K + 2R)c - c^T Rc = \frac{1}{2}c^T Kc. \quad (8.3)$$

The remaining terms are exactly the right side of (8.2). $\square$

Changing R creates no sign provided the square, posterior bracket, and actual payment change together. Changing only a subset has no general sign guarantee, although degenerate fibres may cancel accidentally. In a row split one must still use $\sum_a R_a = R$; the full payment may not be copied into every row.

Combining (8.2) with the R-098 owner identity gives

$$S_R + \frac{1}{2}\mathfrak{C}_{\text{post}} - P_R - \mathbb{E}W_0 = \mathbb{E}\Delta W_R = \mathbb{E}\mathcal{U}_R + \mathbb{E}F_{6.5}. \quad (8.4)$$

After Theorem 7.1, the desired lower form for $F_{6.5}$ is equivalent, up to arbitrarily small total X, Y budgets, to

$$\mathbb{E}\left[S_R + \frac{1}{2}\mathfrak{C}_{\text{post}} - P_R - W_0\right] \geq -\eta\mathbb{E}X - \zeta\mathbb{E}Y - C_{\eta,\zeta}. \quad (8.5)$$

The R-097 bracket-only lower bound is sufficient but strictly stronger because it discards the nonnegative Schur square S_R .

An abstract rank-one Gram shows why an upper form was logically necessary. Let $B(z) = (z^2 - 1)^2$, $U = 1$, $a = -2$, $G = t$, $c = 0$, and $\Gamma = 0$. Then

$$B(1) = B(-1) = B'(1) = 0, \quad B''(1) = 8, \quad L = -16, \quad (8.6)$$

so

$$\mathcal{U}_R = 8t^2, \quad F_{6.5} = -8t^2, \quad (8.7)$$

while the complete endpoint owner vanishes. Thus a lower estimate for $\mathcal{U}_R$ alone cannot control $F_{6.5}$. This is an algebraic recovery fixture, not a production counterexample.

9 Once-only owner ledger and next theorem

The combined result fixes the following order of operations.

1. Inside one root, group the complete (Σ, z) Cartan edge, including the heat compensator, before root differentiation, OU evolution, projection, or squaring.
2. Across roots, use the actual Cameron–Martin cost X once. The later inverse visit does not erase an earlier martingale square, and the terminal mixed payload alone is not a progressive theorem.
3. In control coordinates, use Theorem 5.1 to group all future smoothing shells before squaring. Do not claim that it controls the nonlinear frame.
4. For the frame, retain the mean secant and Jensen residual in the signed cross-Doob pairing. Do not square the secant multiplier separately and do not identify the abstract residual with a production forest without proof.
5. Retain the global Schur mean, low endpoint W_0 , actual matrix payment, q/r means, covariance trace, terminal Θ_R energy, and the R-063 forest exactly once. The forest reconstructs the terminal raw $B_{\text{fr}} : Q$ owner; it is not appended per visit.
6. For the rational row, use the two-sided five-family form and the complete owner target (8.5). Changing R only repartitions its three matched owners.

The next analytic target is therefore not a standalone progressive extension of R-085 (4.11). It is a coefficient-unconditioned, cutoff-uniform lower form for the complete signed production packet in (8.5), simultaneously retaining the linear frame cross term, Jensen/covariance residual, Schur compensation, pure- X allowance, low/heat/trace terms, and the forest. Only after that H_N theorem is proved should REG and the progressive source-union assembly be revisited.

A narrow search of `E:\Dev\Contents` for posterior covariance/bracket, shifted/rational Hessian, Cartan one-use, and root-resampling/Doob statements found deterministic legacy definitions but no reusable theorem that closes this stochastic production packet. This is a repository search result, not a claim about the external literature.

10 Devil’s-advocate review

1. **Objection: Theorem 3.1 proves Cartan (4.11). UPHELD as an overclaim.** The theorem proves exact subdivision invariance within one common root. It supplies neither the spatial atom bound nor its mixed one-use summation.
2. **Objection: terminal $A^* = 0$ cancels the progressive loop. UPHELD as false.** The first root square is nonnegative and already diagonalized; the later fresh root cannot cancel it.
3. **Objection: the no-go also rejects an ηX estimate or R-092. DISMISSED.** Theorem 4.1 excludes only a terminal mixed-only arbitrary- revisit extension. Equation (4.6) records the necessary pure- X coefficient, and R-092’s regular no-revisit class is unchanged.
4. **Objection: Hoeffding multiplicity invalidates R-098. DISMISSED.** The dyadic weights already suppress it. Theorem 5.1 improves exact ownership and signed future-shell grouping.
5. **Objection: Doob–Hardy controls the full nonlinear frame. UPHELD as false.** Equations (6.8) and (6.12) isolate the missing Jensen residual and multiplier moment.
6. **Objection: the Jensen residual is already exactly R-063. UPHELD as unproved.** Only the abstract ordered-reveal identity is proved here; the production ownership identification remains part of the next theorem.
7. **Objection: a favorable ridge R creates positivity. UPHELD as false bookkeeping.** The corresponding square and actual payment cancel the apparent change by (8.2).
8. **Objection: Theorem 7.1 is progressive. UPHELD as an overclaim.** Its authorities prove the absolute bounds only on the fixed-floor regular mutually-orthogonal strict-past one-shot class.

External review should focus on the admissibility of the repeated-range loop, the heat convention in (3.5), the filtration base in (5.4), the distinction between the abstract Jensen residual and production forest, the familywise budget allocation in (7.5), and the payment signs in (8.2).

11 Executable verification

The primary executable uses exact fractions, finite Boolean cubes, random noncommuting positive matrices, and deliberate factor/ownership mutations. The independent executable imports no numerical package or primary module; it recomputes the telescopes, monomial heat contraction, Boolean conditional expectations, Hardy ratios, Jensen/product and spike fixtures, exponent ledger, non-diagonal payment identity, and Gram recovery with standard-library exact arithmetic. The integrated verifier pins the load-bearing authorities, sources, proof note/PDF, both negative results, exploration decisions, public surfaces, and assertion counts.

Run from the repository root:

E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_extended_state_cartan_doob_rational_recovery_verify.py

Result footer

Result ID: A13-CLASSII-EXTENDED-STATE-CARTAN-DOOB-RATIONAL-RECOVERY

Ledger: R-099.

Title: Extended-state Cartan telescope, causal Doob one-use, and
rational recovery boundary.

Precise statement:

Complete value plus heat Cartan edges telescope within a common
root. A genuine progressive reverse visit disproves only a
terminal mixed-only arbitrary-revisit extension. Chronological
Doob differences give an exact one-use Hardy ledger. The full
frame splits into mean secant plus mandatory Jensen residual, and
absolute frame-square transfer fails. The five rational unshifted
families have a two-sided regular one-shot form, and the exact
payment gauge reduces (6.5) to the complete owner lower form.

Scope: Finite cutoff and fixed positive floor. The Cartan telescope uses
one common root and common linear post-operators. The revisit
no-go leaves pure-X payment and R-092 intact. The rational form is
regular mutually-orthogonal strict-past one-shot only.

Dependencies:

R-083; R-084; R-085; R-087; R-092; R-097; R-098.

Evidence grade:

ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command:

E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_extended_state_cartan_doob_rational_recovery_verify.py

Expected output:

Manifest-pinned primary, standard-library independent,
integrated, and aggregate PASS; exit zero.

Negative results:

NG-2026-07-27-A13-PROGRESSIVE-REVISIT-CARTAN-MIXED-PAYLOAD;
NG-2026-07-27-A13-ABSOLUTE-LAST-ROOT-FRAME-TRANSFER.

Falsification gate:

Failure of the extended telescope/heat defect, progressive
scaling, Doob-Hardy/last-root identity, ordered reveal/Jensen
residual, spike, rational exponents/Young estimate, payment gauge,
Gram recovery, authority/hash/PDF/ledger/count, or T4 scope check.

Tier before / after:

T4 / T4; no promotion.

No-overclaim statement:

No production posterior lower form, rational (6.5), complete H_N,
REG, progressive H_A, OVERLAP_src, Nelson, floor removal, measure,
T5--T7, or Sector-A closure is proved.

Next required action:

Prove the coefficient-unconditioned complete signed posterior/
source-action lower form, retaining pure X, linear frame/Jensen,
Schur, low, heat, trace, q/r, payment, and forest exactly once.